

Two certified contiguous data-corner blocks

Conditional multiplication saving $15878574/10^{12}$

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With substantial OpenAI Codex assistance

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Abstract

We refine the data-corner profile in Zhihao Chen’s fixed $(55, 53)$ two-stage construction, PR29. Exact rank-partition identities certify 2265 augmented-minor zeros for contiguous blocks of widths 51 and 45. An exact normalized nonzero-pivot witness and the retained common-basis argument establish their simultaneous availability. The resulting conditional bound is

$$T(n) = O(n(\log n)^{1-\kappa}), \quad \kappa = \frac{15878574}{10^{12}} = 1.5878574 \times 10^{-5} > 2^{-16}.$$

This improves PR29’s reported saving by about 2.205%. All topology, producers, frame end-points, paid copy corrections and analytic mechanisms are inherited unchanged. Exact identity certificates supplement the written proof; the full inherited mathematical theorem is not formally verified.

1 Scope and the actual retained corner

PR29 combines Zhihao Chen’s unequal-axis common basis with Paureel’s two-stage topology and paid correction, icekylinx’s retained-total producers and prescribed bases, and the development route of Swapnil Jain. Its semantic/bulk transfer retains PR23 and RaD/PR20. We credit eumemic and the earlier contributors as recorded in NOTICE. Our contribution is two additional contiguous blocks inside the existing data macro, their weight-independent certificates and the exact assembly. We do not remove any analytic, Gaussian-inverse, uniformity, ordered-affine, exact-recovery or fixed finite-alphabet multitape hypothesis.

Set $a = 55, b = 53, m = ab = 2915, d = a + b - 1 = 107$. The inherited full row/column prescriptions are

$$R_{\text{full}} = [0, \dots, 54; 1, \dots, 54; 0], \quad C_{\text{full}} = [54; 0, \dots, 53; 0, \dots, 54].$$

Take R to be the first 107 entries and C the last 107. Thus

$$R = [0, \dots, 54; 1, \dots, 52], \quad C = [2, \dots, 53; 0, \dots, 54].$$

The row residue is $\beta_i = i \bmod 53$, whereas the column residue is $\gamma_j = (m - d + j) \bmod 53 = (j - 1) \bmod 53$. This offset is essential.

For normalized rank-one pairs $p\xi, v\nu$, the null projector of the data macro $(I - p\xi) \otimes (I - v\nu)$ has actual boundary corner

$$Q_{ij} = \mathbf{1}_{\beta_i=\gamma_j} p_{R_i} \xi_{C_j} + \mathbf{1}_{R_i=C_j} v_{\beta_i} \nu_{\gamma_j} - p_{R_i} \xi_{C_j} v_{\beta_i} \nu_{\gamma_j}.$$

On the nonzero-coordinate open set, divide row i by $p_{R_i}v_{\beta_i}$ and column j by $\xi_{C_j}\nu_{\gamma_j}$. Writing $x_r = (p_r\xi_r)^{-1}$ and $y_t = (v_t\nu_t)^{-1}$ gives

$$M_{ij} = x_{R_i}\mathbf{1}_{R_i=C_j} + y_{\beta_i}\mathbf{1}_{\beta_i=\gamma_j} - 1.$$

There are 109 coefficient coordinates: 55 first-factor vertices, 53 second-factor vertices and a constant $*$. Let f_i have ones at $R_i, \beta_i, *$, and g_j at $C_j, \gamma_j, *$. Exactly

$$M = F \operatorname{diag}(x_0, \dots, x_{54}, y_0, \dots, y_{52}, -1) G^T.$$

Diagonal scaling and the sign relating the actual projector corner to Q preserve all ordered pivot positions and zeros. The normalization equations $\sum 1/x_r = \sum 1/y_t = 1$ are not needed for the zero identities.

2 Exact polynomial-identity certificates

For arbitrary e -row matrices U, V , diagonal D , and coefficient partition $E = X \sqcup Y$,

$$\operatorname{rank}(UDV^T) \leq \operatorname{rank}(U_X) + \operatorname{rank}(V_Y).$$

Split the product into its X and Y summands and bound the first by its left factor and the second by its right factor. If this sum is below e , its determinant vanishes for all diagonal weights. For an invertible selected prefix minor P , each next Schur entry is the corresponding augmented determinant divided by $\det P$, with row and column order retained.

The claimed rightmost-pivot sequence is

$$\begin{aligned} &(0, 106); \quad (i, i + 54), \quad 1 \leq i \leq 51; \\ &(52, 54), (53, 53), (54, 52), (55, 51), (56, 50), (57, 49); \\ &\quad (i, i - 54), \quad 58 \leq i \leq 102; \\ &(103, 3), (104, 2), (105, 1), (106, 0). \end{aligned}$$

Outside the two long runs each pivot uses the largest available column. For every still available column to the right of a claimed long-run pivot, the supplied certificate gives a partition with rank sum strictly below the augmented minor size. There are $51 \cdot 50/2 + 45 \cdot 44/2 = 2265$ such minors. Bits 0–54 encode first-factor vertices, bits 55–107 second-factor vertices and bit 108 the constant. The prefix rows and columns follow the displayed order.

The search uses matroid intersection and an incidence-forest rank oracle to find partitions. The mathematical implication does not depend on that search: all 4530 rank values are checked directly by integer elimination with gcd reduction. A separate verifier independently reconstructs the required minor set and verifies its exact coverage, both ranks and the strict deficiency. Thus these are finite exact polynomial-identity certificates for the fixed geometry, not numerical sampling of a conjectured zero pattern.

The search oracle itself has a simple proof. For a vertex-column subset S of a forest incidence matrix, rank is $|S|$ minus the number of components wholly contained in S , including isolated vertices. Adding the constant column increases rank exactly when some component has omitted vertices on both bipartition sides. On a connected tree, solutions to $z_R + z_B = 1$ have $z_R = t, z_B = 1 - t$; the omitted coordinates can all be zero precisely when they do not occur on both sides. The direct integer checks independently verify every oracle value used in a final certificate.

3 Nonzero minors and one simultaneous basis

Take the exact normalized rational witness

$$p_r = v_t = 1, \quad \xi_r = \frac{(r+1)^2}{\sum_{s=1}^{55} s^2}, \quad \nu_t = \frac{(t+1)^3}{\sum_{s=1}^{53} s^3}.$$

All coordinates are nonzero and both normalizations hold. The checker reconstructs the 107-square matrix over $\mathbb{Q}$, performs rightmost-nonzero elimination and compares all 107 exact pivot values with the saved witness. Hence every required prefix minor is a nonzero rational function on the normalized projector family. The resulting run lengths are

$$1, 51, 1, 1, 1, 1, 1, 1, 45, 1, 1, 1, 1.$$

The retained permutation completions leave L_a, L_b free. Any actual rank-one projector is conjugate by $\mathrm{GL}_a(\mathbb{Q})$ to any normalized rank-one pair, and independently for the second factor. For each physical data pair, the displayed witness therefore establishes nonvanishing in the same $\mathrm{GL}_{55} \times \mathrm{GL}_{53}$ family as the inherited local and auxiliary minors. After clearing denominators, the product over finitely many data pairs, all prior minors and basis determinants is nonzero on this irreducible rational family. A common rational basis avoiding all zeros exists. We neither select a different basis per gate nor assume that the displayed witness satisfies every physical pair simultaneously. The zero identities hold for all weights and impose no conflicting closed specialization. The inherited address-prime choice avoids finitely many denominators and minors. Reverse invocation has the same positive residual profile.

4 Physical contiguity and exact assembly

The width-51 run uses rows 1–51 and physical columns 2863–2913. The width-45 run uses rows 58–102 and columns 2812–2856. The untouched large-projector middle is 107–2807, width 2701. These intervals are disjoint; the inherited ordered-climb lemma charges each run as one child interchange without gathering or reordering. The data profile is

$$11 \text{ singletons} + [51, 45, 2701], \quad 11 + 51 + 45 + 2701 = 2808 = m - d.$$

Across the existing $2N$ data macros, remove $192N$ singleton calls and add $2N$ each of widths 51 and 45. The physical W , total rank mass, largest child, auxiliary profiles and paid N rank-one endpoint-copy corrections are unchanged.

The full child histogram is reconstructed from pinned PR29 and this exact delta. The rational log/exponential enclosure supports $a_b = 15879079/10^{12}$. It uses positive logarithm tails and

$$e^u \leq 1 + u + \frac{u^2}{2(1-u/3)}, \quad 0 \leq u < 1.$$

The next bit grid point fails this sufficient enclosure; no impossibility or optimality is inferred. The complex saving remains $18/10^6$, $\beta = 1/10$, guard $C_1 = 1$ and joint row degree 47000. The entire inherited finite bridge is unchanged.

For $h = 10^{-8}$ put $q = a_b(1-2h)$, $c = q(1+h)$, $\epsilon = (1-h)/(1+c+q)$, $\lambda' = 1-q$, $\lambda = (1-a_b+\lambda')/2$, $r = (\epsilon q + 1 - \epsilon)/2$ and $\delta = h/8$. The parameterized inherited assembly checks all 47 strict constraints and seven margins at $\kappa = 15878574/10^{12}$, with their eventual cutoffs. Negative controls omit the data blocks, use a wrong residue/identity certificate, restore obsolete guard or exposure budgets, and inflate κ to its next grid point.

5 Reproduction and review boundary

Run `make two-stage-corners-certificate` and `make two-stage-corners-patch`, followed by the focused `test_two_stage_corners.py` suite. The identity-search replay is separately available through `two-stage-corners-identities`. Full `make verify` reruns inherited checks and the fresh PR29 producer replays. The validation receipt distinguishes focused checks from completion of that full rerun.

The focused patch applies to PR29 commit9d96327, not the original OpenAI release. All inherited sources remain byte-identical. This proof, exact identities, independent checks and refinement are by Rohan Arun with substantial OpenAI Codex assistance; the inherited topology, producer, basis and analytic mechanisms retain their original attributions. No practical speedup or global-optimality claim is made.